

KARA-SUU, Kyrgyzstan

WMO: 386160

Lat: 40.700N

Lon: 72.900E

Elev: 2848

StdP: 13.25

Time Zone: 6.00 (E06)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.4	15.6	4.7	7.8	13.5	8.8	9.6	16.8	11.6	38.4	8.8	31.9	1.8	140	0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.0	94.3	68.5	91.7	67.6	89.6	66.8	71.2	88.3	69.7	87.2	68.4	85.8	2.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	106.6	79.8	63.7	98.3	79.1	62.0	92.5	78.4	37.1	88.1	35.7	87.6	34.5	86.0	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	9.0	7.0	DB	6.6	99.7	6.0	3.5	2.3	102.2	-1.2	104.3	-4.5	106.2	-8.8	108.7	(4)
(5)			WB	6.0	74.6	5.6	2.8	2.0	76.7	-1.3	78.3	-4.4	79.9	-8.5	82.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.9	30.9	35.2	47.7	58.2	66.6	74.4	78.1	75.9	68.5	56.4	44.0	33.4		
	DBStd	17.78	7.56	8.30	7.96	7.22	5.61	5.00	3.93	4.20	5.48	7.08	7.44	6.83		
	HDD50	1910	593	417	136	17	1	0	0	0	0	25	207	514		
	HDD65	4616	1058	835	537	220	48	3	0	1	29	277	629	979		
	CDD50	4056	0	2	65	264	515	731	872	803	554	223	27	0		
	CDD65	1284	0	0	0	17	96	284	407	338	133	9	0	0		
	CDH74	13307	0	0	1	156	923	2961	4442	3440	1279	103	2	0		
	CDH80	5546	0	0	0	20	247	1253	2121	1519	373	13	0	0		
(14) Wind	WSAvg	2.4	1.9	2.4	2.8	2.9	2.8	3.0	2.6	2.5	2.4	2.1	2.0	1.9		
(15) Precipitation	PrecAvg	15.8	1.3	1.7	2.1	2.0	2.0	0.9	0.5	0.3	0.3	1.1	1.8	1.8		
	PrecMax	25.4	4.5	3.5	4.4	5.7	5.6	2.8	1.1	1.1	0.7	4.7	6.0	3.7		
	PrecMin	8.9	0.1	0.4	0.5	0.7	0.1	0.1	0.0	0.0	0.0	0.0	0.2	0.4		
	PrecStd	3.8	0.9	0.8	1.1	1.0	1.4	0.8	0.4	0.3	0.2	1.0	1.5	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.7	61.3	72.9	82.9	88.9	96.1	97.7	95.1	88.8	81.5	70.8	53.5		
		MCWB	43.0	48.3	55.3	61.7	65.2	68.6	69.0	69.7	64.9	59.5	53.6	43.9		
	2%	DB	46.5	53.1	68.3	78.4	85.7	92.6	95.1	92.4	86.8	76.7	63.5	48.7		
		MCWB	40.2	44.3	53.4	59.8	64.4	67.2	68.7	68.3	63.9	58.0	50.9	41.7		
	5%	DB	42.9	49.3	63.9	74.7	82.7	89.9	92.6	90.3	84.5	73.0	58.9	45.3		
		MCWB	37.9	42.6	51.6	58.4	63.5	65.8	68.0	67.3	63.4	56.0	48.3	39.4		
	10%	DB	40.3	46.0	60.0	71.1	79.5	87.2	90.1	88.2	81.6	69.3	55.2	42.2		
		MCWB	36.1	40.7	49.7	57.0	62.1	65.2	67.4	66.5	62.0	54.3	46.4	37.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.3	49.4	58.2	64.9	68.4	72.7	73.9	72.9	69.0	61.3	55.4	45.9		
		MCDB	50.9	59.4	67.6	77.5	83.9	89.5	90.5	90.0	84.9	78.0	68.2	51.0		
	2%	WB	40.9	45.4	54.7	62.1	66.3	69.6	71.3	70.8	66.4	59.1	52.0	42.4		
		MCDB	45.1	50.9	65.1	74.1	81.9	87.0	88.8	87.5	83.2	73.6	60.9	47.0		
	5%	WB	38.5	43.1	52.4	60.0	64.7	67.7	69.9	69.3	64.4	57.1	49.5	40.2		
		MCDB	42.2	48.4	61.5	71.5	79.2	85.7	88.1	86.0	80.9	70.2	57.2	44.4		
	10%	WB	36.5	41.0	50.5	58.2	63.2	66.2	68.7	68.0	62.7	55.4	47.4	38.1		
		MCDB	39.7	45.5	58.7	68.9	76.8	83.9	86.6	84.7	79.2	66.5	54.2	41.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	13.2	16.1	18.1	20.7	23.2	23.0	22.8	22.6	19.9	15.4	12.6		
		MCDBR	15.3	18.0	22.2	23.6	25.7	26.7	26.2	24.8	25.3	24.3	21.3	16.8		
		MCWBR	10.7	11.3	11.6	12.0	11.9	11.2	11.2	10.7	11.8	11.7	11.8	11.5		
		MCDBR	13.8	16.3	18.8	20.2	23.0	24.1	23.8	23.2	23.8	21.6	18.8	15.2		
	5% WB	MCWBR	10.2	11.0	10.8	11.3	11.9	11.5	11.5	11.1	12.2	11.0	10.8	11.0		
(40) Clear-Sky Solar Irradiance	taub		0.337	0.383	0.452	0.477	0.467	0.468	0.454	0.422	0.412	0.439	0.398	0.336		
	taud		2.224	2.109	1.961	1.941	2.029	2.046	2.090	2.182	2.141	1.984	2.057	2.215		
	Ebn at Noon		253	256	249	253	259	259	261	266	260	231	223	241		
	Edh at Noon		31	40	53	57	54	53	50	44	43	45	36	29		
(44) All-Sky Solar Radiation	RadAvg		707	985	1334	1646	1986	2268	2309	2122	1742	1134	747	603		
	RadStd		51	76	97	126	129	101	96	90	69	115	95	55		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	+0.92	N/A	N/A	N/A	N/A	+135	+115

Nomenclature: See separate page